KFParticle Λ–^{4}He Correlation Analysis: QA Summary

STAR analysis working report

2026-10-03 19:36 JST

1 Sample and scope

Au+Au fixed-target sample, beam-energy label 13.5 GeV; production 2020 P24iy. This beam label is *not* a claim that $\sqrt{s_{NN}}$ equals this value. **Partial merge:** 542/545 subjobs are included; 3 missing or excluded subjobs are not represented. No missing yield is filled in.

ROOT: `auau13p5_anaFemtoLambda_4He_KFParticle_2020P24iy_partial_merge.root`

SUMS ID: `1213058CE62BCF113B2E3C75ACBD7053`

This report reuses the structure of the earlier tools summaries, but reads only the KFParticle merged ROOT above. Legacy ROOTs, legacy PDFs and selection cuts are unchanged. The methods below are post-processing cross-checks, not new production selection or a final physics result.

Measurement	Stored value
Input-event counter	36,004,989
Reconstructed-event counter	34,883,141
Selected-Λ counter	3,044,144
Signal-window candidate counts	2,463,680
Width-scaled sideband background	50,103
Inclusive purity proxy P	0.979663 ± 0.000055

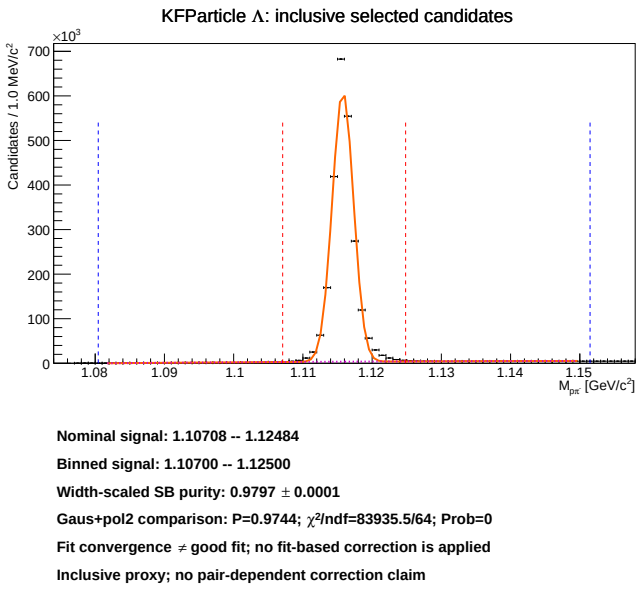


Figure 1: Inclusive $p\pi^-$ invariant mass. Vertical markers show the unchanged production signal and sideband gates; the fit is a diagnostic only.

2 Mass gates and purity convention

The Maker gate parameters are $\mu = 1.11596$ GeV/c^2, $\sigma = 0.00296$ GeV/c^2, with signal $|M - \mu| < 3\sigma$. The left/right sidebands extend from 3σ to 12σ. These fixed gate parameters are not refitted to redefine the event selection.

Region	Nominal edges	Integrated bin edges
Signal	1.10708–1.12484	1.107–1.125
Left SB	1.08044–1.10708	1.08–1.107
Right SB	1.12484–1.15148	1.125–1.151

All mass edges are in GeV/c^2. For the inclusive diagnostic, whole bins with centers inside each nominal interval are integrated, without double counting boundary bins. The binned diagnostic therefore only approximates the unbinned Maker gate.

$$P_{\text{inclusive}} = 1 - \alpha_{\text{bins}} \frac{N_L + N_R}{N_W}, \qquad \alpha_{\text{bins}} = \frac{\Delta M_W}{\Delta M_L + \Delta M_R}. \tag{1}$$

Here $\alpha_{\text{bins}} = 0.339623$, whereas the nominal sideband scale used with the stored pair histograms is $\alpha = 0.333333$. The estimate assumes a locally flat background across the sidebands; its displayed error is statistical and excludes model uncertainty. **CFsub uses the measured value $P = 0.979663$ above, not an inherited fixed S/N.** This is an inclusive-candidate proxy, not a measured pair-, centrality- or k^*-dependent purity. Its uncertainty is not propagated through the legacy constant-purity CFsub curves; those errors must not be read as total uncertainties. The current left and right sidebands contain 29,287 and 118,239 candidates. Their right/left count-density ratio is 4.19253. This measured asymmetry warns against treating the flat-background proxy as a precision purity determination; its small statistical error does not cover interpolation or shape uncertainty. The independent Gaussian-plus-quadratic fit has status 0, covariance status 2, $\chi^2/\text{ndf} = 83935.5/64$, and fit probability 0. Its comparison purity is 0.974422. This is a diagnostic value only, not adopted for CF correction. Numerical convergence does not establish model adequacy: a very small probability or large χ^2/ndf indicates that the simple shape model is inadequate. The same limitation must be considered for the projection fits in Methods 3 and 5.

3 Common post-processing choices

The comparison macros use k^* rebinning by 4 (20 MeV/c bins) and normalize SE/ME in $0.4 < k^* < 0.6$ GeV/c. The displayed k^* range is 0–0.4 GeV/c; normalization still uses the full stated interval outside the displayed range. These are summary post-processing choices, distinct from the Maker CF settings. Raw histograms are summed *before* a grouped CF is formed. Method 5 keeps $W = 2$ neighboring bins on each side of the central bin: the nominal interior window spans 100 MeV/c. Therefore changing the displayed/analysis bin width also changes this physical smoothing width, although W itself is unchanged. Edge windows are truncated. Two group sets are used consistently: (20–80%, 0–20%) and (0–10%, 10–20%, 20–60%). In the first group-set plots below, the left panel is 20–80% and the right panel is 0–20%. Fine-bin plots show all nine centralities, from 70–80% to 0–5%.

4 Definitions and limitations

Let W denote the signal mass gate, $L + R$ the summed sidebands, and $\mathcal{N}$ the normalization obtained in the stated k^* interval. For raw sideband subtraction,

$$C_{\mathrm{SBsub}}(k^*) = \mathcal{N}_{\mathrm{sub}} \frac{N^W_{\mathrm{SE}} - \alpha(N^L_{\mathrm{SE}} + N^R_{\mathrm{SE}})}{N^W_{\mathrm{ME}} - \alpha(N^L_{\mathrm{ME}} + N^R_{\mathrm{ME}})}. \tag{2}$$

For constant-purity CF subtraction, signal and summed-sideband CFs are normalized separately:

$$C_{\mathrm{CFsub}}(k^*) = \frac{C_W(k^*) - (1 - P)C_{L+R}(k^*)}{P}. \tag{3}$$

The two sidebands are added at the count level; C_{L+R} is not the simple average of independently normalized left and right CFs. The methods assume that the sideband background represents the signal-region background, in count shape or correlation shape respectively. Neither assumption is validated by plotting alone. Method 3 fits each mass projection with a Gaussian plus quadratic background, separately for SE and ME, and constructs a normalized ratio of fitted signal yields. Method 5 estimates the purity using overlapping neighboring k^* windows and applies it to the central bin. Overlap introduces covariance between adjacent estimates; the legacy diagonal-error treatment does not include that covariance.

- Statistical errors do not include CF-normalization uncertainty, background-model systematics, or the constant inclusive-purity uncertainty.
- Direct subtraction can produce small or nonpositive denominators. Sparse or failed mass fits do not constitute zero physical signal. Review fit audit logs and invalid-bin diagnostics before using any plotted point quantitatively.
- Whole mass bins used in projection fits need not match the exact stored signal-gate pair histograms at the boundary. A method difference can therefore include bin-edge effects.
- Missing subjobs and possible coverage biases are not corrected. This report does not compare efficiencies or infer purity for the missing sample.
- No method is declared superior from smoothness alone, and no source radius is established as a final physical result.

5 Raw sideband subtraction: grouped spectra

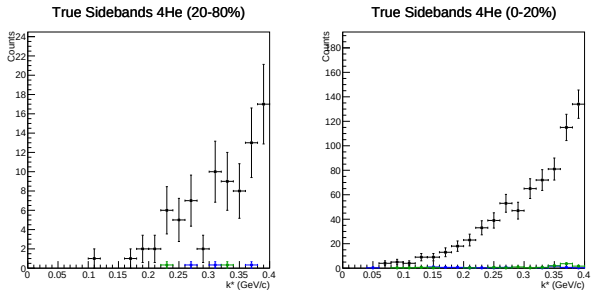


Figure 2: SE signal-gate and sideband distributions.

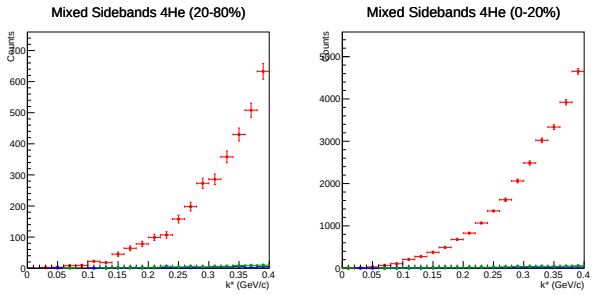


Figure 3: ME signal-gate and sideband distributions.

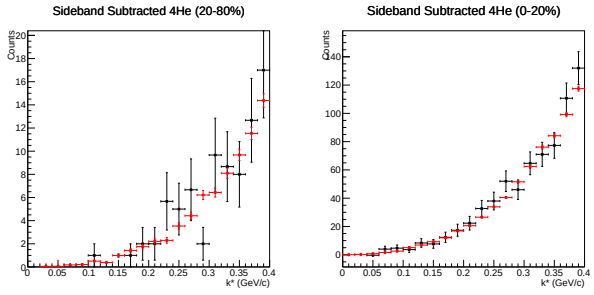


Figure 4: Background-subtracted SE and ME spectra.

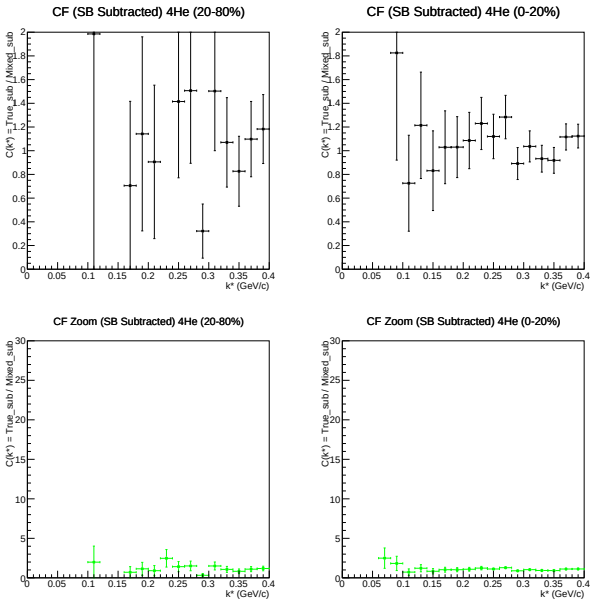


Figure 5: Normalized CF after raw sideband subtraction.

6 Constant-purity CF subtraction

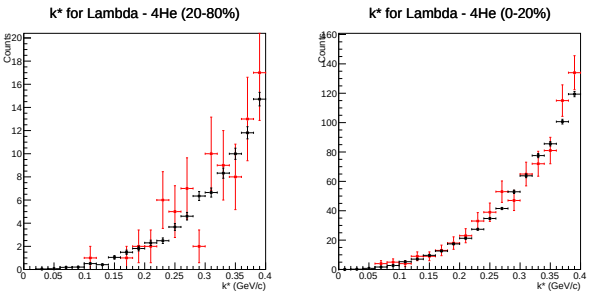


Figure 6: SE and ME in the signal mass gate.

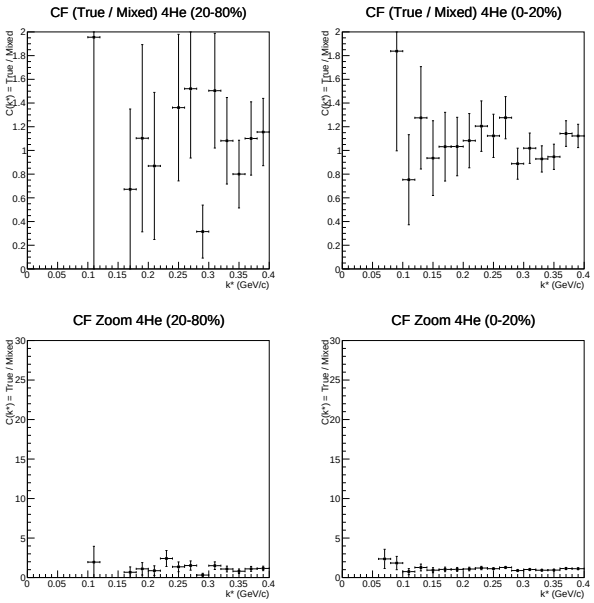


Figure 7: Uncorrected signal-gate CF.

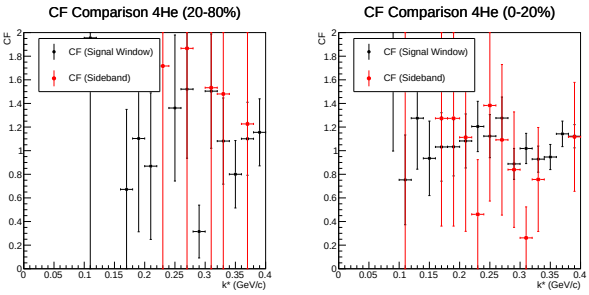


Figure 8: Signal-gate CF and summed-sideband CF.

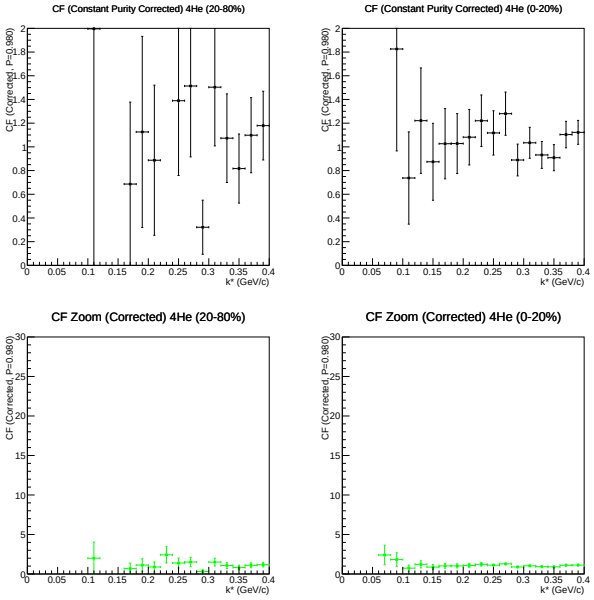


Figure 9: CFsub using the current inclusive-purity proxy.

7 Method 3: mass-fit signal yields

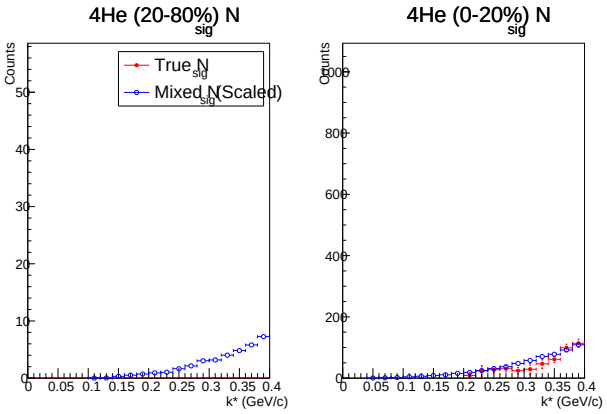


Figure 10: Mass-fit SE and ME signal yields.

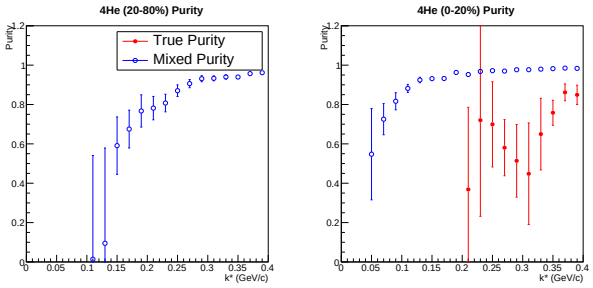


Figure 11: SE and ME fitted purity versus k^*.

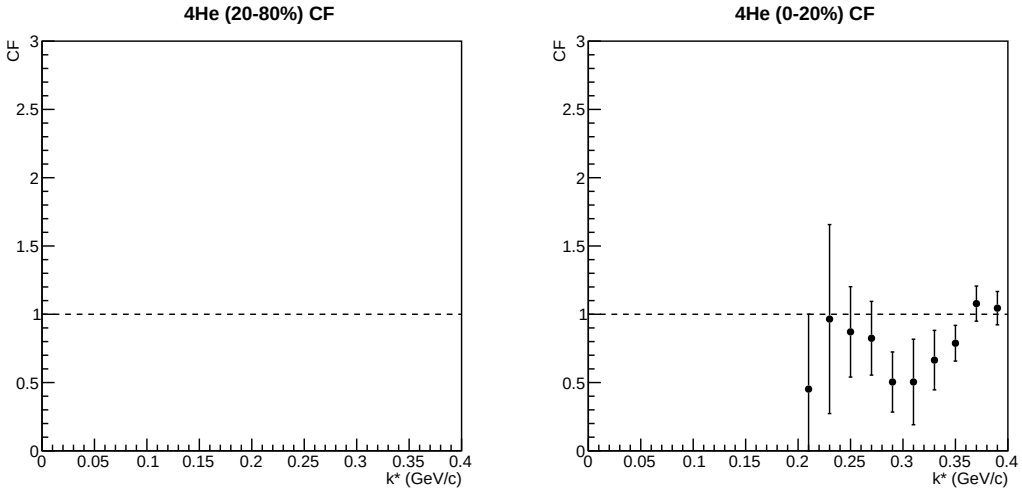


Figure 12: Normalized fitted-signal-yield ratio. Fit quality must be checked independently.

8 Method 5: sliding-window comparison

Both fine-centrality figures below show all nine centralities. The 0–5% bin is the bottom-right panel; page 2 is a purity summary, not a single-centrality page.

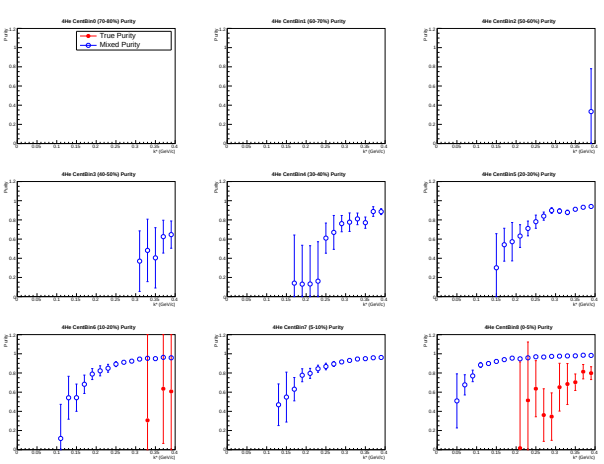


Figure 13: Method 3 purity in nine centrality bins.

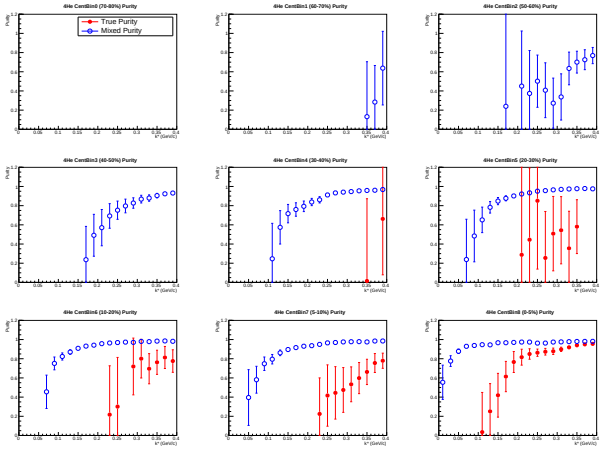


Figure 14: Method 5 sliding-window purity in the same bins.

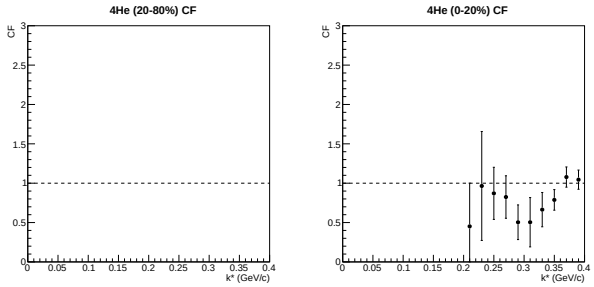


Figure 15: Method 3 grouped CF.

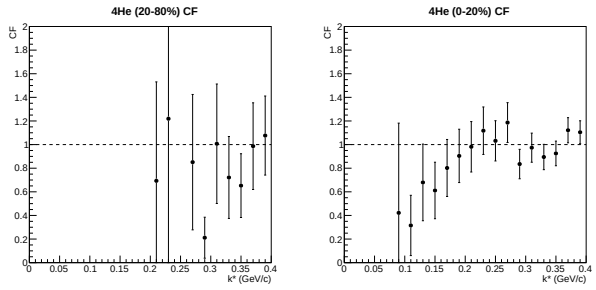


Figure 16: Method 5 grouped CF; adjacent-bin covariance is omitted.

9 Exploratory Lednicky–Lyuboshits fits

The blue fits hold $a_0 = 4.01$ fm and $r_{\mathrm{eff}} = 1.84$ fm fixed and fit R and a normalization. The red fits also vary a_0 and r_{eff}. These reproduce the existing model/sign convention, not an independently validated interaction extraction. The fit interval is $0 < k^* < 0.4$ GeV/c: only about 20 bins of width 20 MeV/c are available before invalid-bin exclusions. Four free parameters can therefore be weakly constrained. Parameter bounds, covariance quality and fit status matter in addition to the displayed χ^2/ndf.

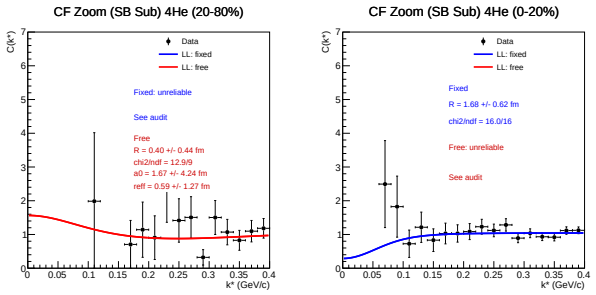


Figure 17: LL fits to raw-SB-subtracted CFs.

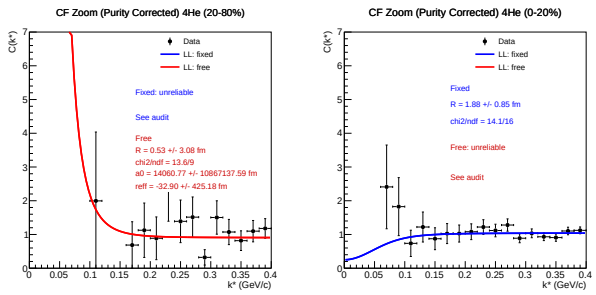


Figure 18: LL fits to constant-purity CFsub CFs.

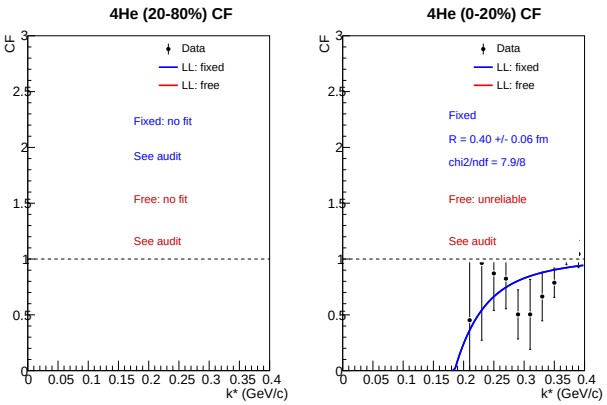


Figure 19: LL fits to Method 3 CFs.

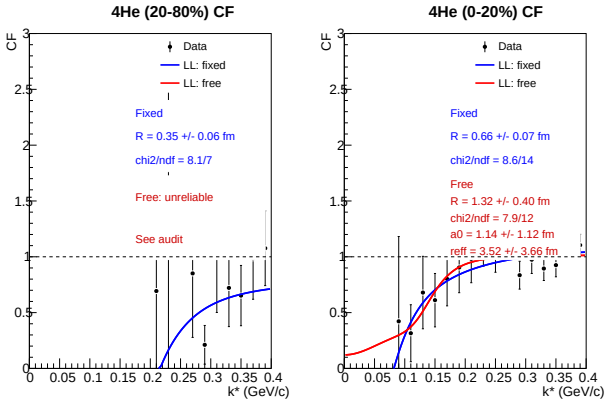


Figure 20: LL fits to Method 5 CFs; window covariance is not fitted.

10 Additional centrality groups

The following pages use the same groups for every method: 0–10%, 10–20% and 20–60%. This corrects the inconsistent third-group definitions in some legacy macros.

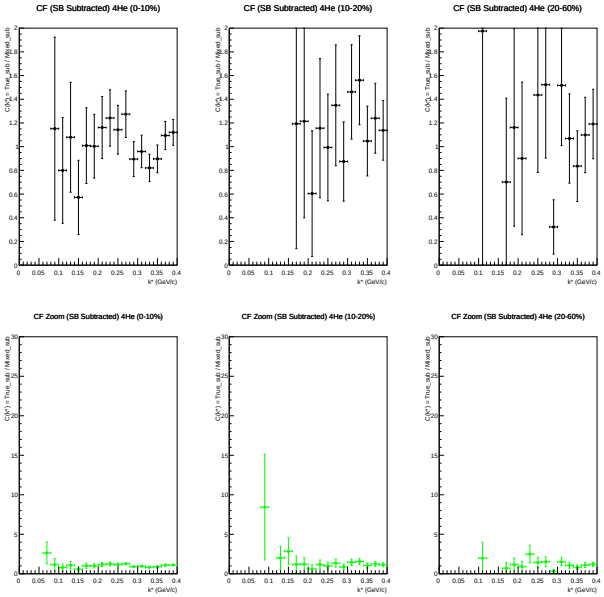


Figure 21: SBsub, second group set.

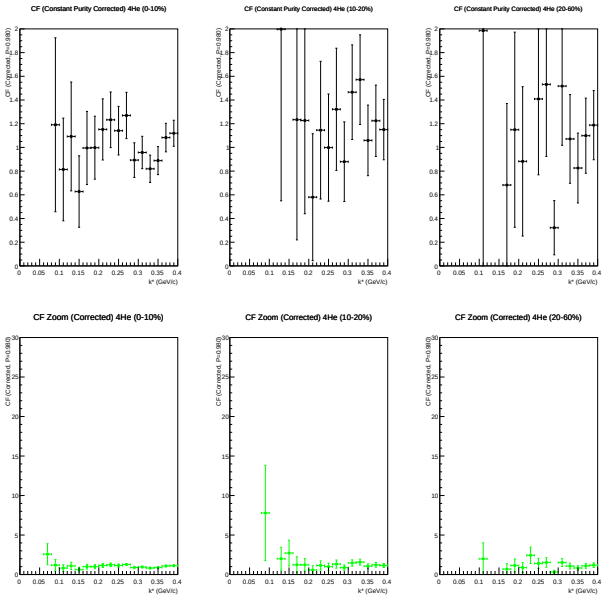


Figure 22: Constant-purity CFsub, second group set.

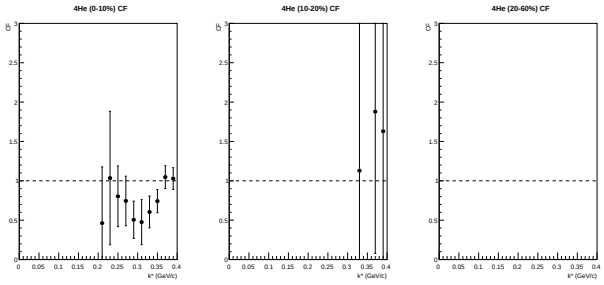


Figure 23: Method 3, second group set.

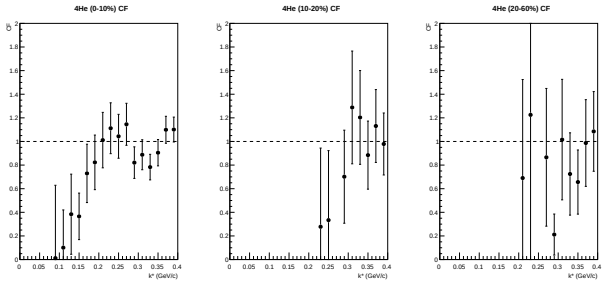


Figure 24: Method 5, second group set.

11 LL fit audit and reproducibility

Fit status 0 alone does not establish model adequacy. Covariance status and boundary flags are retained separately below. Repeated optimizer calls are reduced to the last result for each fit object. Full parameters and diagnostics remain in the TSV files.

Mass-fit macro	Total	Accepted	Boundary	Rejected
draw_sum_CFsub_method3	500	396	65	104
draw_sum_CFsub_method5	500	451	40	49
draw_CFsub_method3	900	426	92	474
draw_CFsub_method5	900	561	77	339
Fit_sum_CFsub_method3	500	396	65	104
Fit_sum_CFsub_method5	500	451	40	49

Accepted here means only the implemented numerical checks passed; it is not a goodness-of-fit or physics validation. Boundary cases are included in accepted counts and are shown separately as warnings. For object `fit1_s_g`, a_0 and r_{eff} are fixed; `fit2_s_g` is the all-free fit. The zero-based group-set index $s = 0$ means (20–80%, 0–20%) and $s = 1$ means (0–10%, 10–20%, 20–60%); g is the position in that list.

Macro / fit	Object	Status	Cov.	Bound	R (fm)
SBsub	fit1_0_0	0	3	1	6*
SBsub	fit1_0_1	0	3	0	1.679
SBsub	fit1_1_0	0	3	0	1.577
SBsub	fit1_1_1	0	3	1	6*
SBsub	fit1_1_2	0	3	1	6*
SBsub	fit2_0_0	0	3	0	0.4018
SBsub	fit2_0_1	0	3	1	6*
SBsub	fit2_1_0	0	3	0	5.999
SBsub	fit2_1_1	0	3	0	0.5727
SBsub	fit2_1_2	0	3	0	0.512
CFsub	fit1_0_0	0	3	1	6*
CFsub	fit1_0_1	0	3	0	1.879
CFsub	fit1_1_0	0	3	0	1.71
CFsub	fit1_1_1	0	3	1	6*
CFsub	fit1_1_2	0	3	1	6*
CFsub	fit2_0_0	0	3	0	0.527
CFsub	fit2_0_1	0	3	1	6*
CFsub	fit2_1_0	0	3	1	6*
CFsub	fit2_1_1	0	3	0	0.5757
CFsub	fit2_1_2	0	3	0	0.5294
CFsub_method3	fit1_0_0	-1	-1	0	not fitted
CFsub_method3	fit1_0_1	0	3	0	0.3987
CFsub_method3	fit1_1_0	0	3	0	0.3843
CFsub_method3	fit1_1_1	0	3	0	0.289
CFsub_method3	fit1_1_2	-1	-1	0	not fitted
CFsub_method3	fit2_0_0	-1	-1	0	not fitted
CFsub_method3	fit2_0_1	4	2	1	0.1*
CFsub_method3	fit2_1_0	0	3	0	0.7155
CFsub_method3	fit2_1_1	-1	-1	0	not fitted
CFsub_method3	fit2_1_2	-1	-1	0	not fitted
CFsub_method5	fit1_0_0	0	3	0	0.3484
CFsub_method5	fit1_0_1	0	3	0	0.659
CFsub_method5	fit1_1_0	0	3	0	0.5923
CFsub_method5	fit1_1_1	0	3	0	0.3467
CFsub_method5	fit1_1_2	0	3	0	0.3484
CFsub_method5	fit2_0_0	0	3	1	0.1*
CFsub_method5	fit2_0_1	0	3	0	1.322
CFsub_method5	fit2_1_0	0	3	0	1.206
CFsub_method5	fit2_1_1	0	3	0	0.3265
CFsub_method5	fit2_1_2	0	2	0	0.1023

An asterisk marks an unreliable optimizer result. Not fitted suppresses initial parameters left behind when too few valid points were available. Even unmarked fits remain exploratory and omit the covariance sources noted above. **Reproduction records:** Measurements: `inspect.json`; macros/settings: `macro_manifest.json`; input/configuration: `run_state.json`; fit diagnostics: `logs/`; figure hashes/pages: `report_metadata.json`. Commands and full figure index: `README.md`.